

TokTickIT Lab 2 — Final Engineering Deliverable & Quality Assurance Report

Project: TokTickIT — Spec-Driven IT Support Ticket Management System (Lab 2) **Author / Team:** chanya06
Reviewer: Peer Reviewer (lmaybelgracel) **Target Branch:** lab2-staging -> main **Date:** September 5, 2026

Answer Part 1: Project Architecture & Spec-Driven Development (Spec DD) Overview

1.1 Methodology

TokTickIT Lab 2 was engineered using **Spec-Driven Development (Spec DD)**. Before any source code implementation was initiated, four foundational specification contracts were written and committed under docs/lab-02/ : 1. specification.md : High-level system requirements, Business Rules (BR-01..BR-16), Functional Requirements (FR-01..FR-15), and Acceptance Criteria (AC-01..AC-10). 2. api-spec.md : OpenAPI/REST contract endpoints, request/response schemas, validation rules, HTTP status codes (200 , 201 , 400 , 403 , 404 , 413 , 422 , 500), and header isolation (X-Requester-Id). 3. ui-spec.md : User interface layouts, Zen Green design system color palette tokens (#006B3C , #0B7A46 , #EAF6EF , #F5F7F6), responsive grid breakpoints (Desktop 1280px, Tablet 768px, Mobile 375px), and error messaging placement. 4. tests.md : Test strategy, planned automated test cases across 5 levels, traceability matrix, visual inspection checklist, and known limitations.

1.2 Full-Stack Architecture Stack

- **Frontend (Client):** React 18, TypeScript, Vite, Bootstrap 5 (Custom Zen Green styling), Context API for Development Requester context state management. - **Backend (Server):** Node.js, Express, TypeScript, Prisma ORM, Multer for multipart file uploads. - **Database:** PostgreSQL with Prisma Schema models. - **Testing Engine:** Vitest & Supertest (Server API & Unit), Vitest & React Testing Library (Client Components & Style), Playwright (End-to-End Browser Journeys & Responsive Screenshot Captures).

Answer Part 2: Database Prisma Schema & Migration Strategy

2.1 Database Schema Models

The relational database models defined in server/prisma/schema.prisma support multi-tenant identity testing, ticket creation, and attachment lifecycles:

```

model DevelopmentRequester {
  id      Int      @id @default(autoincrement())
  name    String
  email   String   @unique
  department String?
  isActive Boolean @default(true)
  tickets Ticket[]
  createdAt DateTime @default(now())
  updatedAt DateTime @updatedAt
}

model Category {
  id      Int      @id @default(autoincrement())
  name    String   @unique
  isActive Boolean @default(true)
  tickets Ticket[]
  createdAt DateTime @default(now())
}

model RelatedSystem {
  id      Int      @id @default(autoincrement())
  name    String   @unique
  description String?
  isActive Boolean @default(true)
  tickets Ticket[]
  createdAt DateTime @default(now())
}

enum RequestedPriority {
  LOW
  MEDIUM
  HIGH
  URGENT
}

enum TicketStatus {
  NEW
  OPEN
  IN_PROGRESS
  PENDING
  RESOLVED
  CLOSED
}

model Ticket {
  id          Int          @id @default(autoincrement())
  ticketNumber String      @unique
  requesterId Int
  categoryId  Int
  relatedSystemId Int
  summary     String
  description  String
  requestedPriority RequestedPriority @default(MEDIUM)
  itPriority   RequestedPriority @default(MEDIUM)
}

```

```

status      TicketStatus    @default(NEW)
ticketOwner  String?
createdAt   DateTime        @default(now())
updatedAt   DateTime        @updatedAt
requester    DevelopmentRequester @relation(fields: [requesterId], references: [id])
category     Category       @relation(fields: [categoryId], references: [id])
relatedSystem RelatedSystem  @relation(fields: [relatedSystemId], references: [id])
attachments  Attachment[]
}

model Attachment {
  id          Int    @id @default(autoincrement())
  ticketId    Int
  filename     String
  originalName String
  mimeType     String
  sizeBytes    Int
  filepath     String
  isRemoved    Boolean @default(false)
  removalReason String?
  removedAt    DateTime?
  createdAt    DateTime @default(now())
  ticket       Ticket  @relation(fields: [ticketId], references: [id], onDelete: Cascade)
}

```

2.2 Idempotent Seeding Strategy

`server/prisma/seed.ts` uses Prisma `upsert` operations for categories, related systems, and development requesters. Executing `npx prisma db seed` repeatedly runs cleanly without key duplication or constraint violations.

Answer Part 3: Identity & Requester Isolation Security Model

3.1 Context & Isolation

In Lab 2, real user authentication is out of scope (`Known Limitations`). To simulate multi-tenant data isolation, a **Development Requester Selector** component (`RequesterSelectorModal.tsx`) allows switching the active requester context (e.g. Jennifer Anderson vs Michael Brown).

3.2 Authorization & Conflict Middleware

- **Header Synchronization:** The client passes the active requester ID in HTTP header `X-Requester-Id` and URL query parameter `requesterId` . - **Query / Header Conflict Verification:** The server verifies that if both `X-Requester-Id` header and `requesterId` query parameter are present, their values must match exactly; otherwise, the server rejects the request with `400 Bad Request` . - **Cross-Requester Ownership Guard (`403 Forbidden`):** Every API endpoint (`GET /api/tickets` , `GET /api/tickets/:id` , `GET /api/tickets/:id/attachments` , `POST /api/tickets/:id/attachments` , `GET /api/attachments/:id/download` , `DELETE /api/attachments/:id/soft-remove`) verifies

that the requested ticket or attachment belongs to the active `requesterId`. Cross-requester access attempts return `403 Forbidden`.

Answer Part 4: Ticket Creation & Sequential Numbering Engine

4.1 Ticket Number Format & Generation

Tickets are assigned a unique, immutable ticket number formatted as `TKT-YYYY-XXXXXX` (e.g. `TKT-2026-000001`). Sequence numbers reset yearly.

4.2 Initial File Attachments & Transaction Compensation (`BR-15`)

- `POST /api/tickets` handles multipart/form-data with up to 5 initial file attachments. - **Transaction Compensation Strategy:** - File binary signature (magic bytes), extension, size ($\leq 5\text{MB}$), and MIME types are validated before database persistence. - Ticket record creation and initial attachment records creation are wrapped in a Prisma `$transaction`. - If any validation error, DB constraint error, or server exception occurs, all uploaded files are unlinked/deleted from disk immediately (`fs.unlinkSync`), guaranteeing zero orphan files on server storage.

Answer Part 5: My Tickets Search, Multi-Select Filter & Pagination Engine

5.1 Endpoint Capabilities (`GET /api/tickets`)

- **Debounced Search:** Searches across `ticketNumber` and `summary` with 300ms debounce. - **Multi-Select Filtering:** Category (`categoryId`), Requested Priority (`requestedPriority`), and Status (`status`) accept comma-separated strings or arrays. - **Sorting:** Supports `createdAt` and `ticketNumber` sorting (`asc` / `desc`). - **Pagination:** Returns paginated payload containing `data` list with `attachmentCount` metadata and `pagination` details (`totalItems`, `totalPages`, `currentPage`, `pageSize`).

Answer Part 6: Ticket Detail & Ownership Authorization Guard

6.1 Ticket Detail Display (`GET /api/tickets/:id`)

- Displays read-only ticket information: Ticket Number, Created Date / Ticket Date, Category, Related System, Priority, Status, Summary, Description, and Attachment Section. - Enforces ownership check (`403 Forbidden` if requester ID does not match ticket owner).

Answer Part 7: Attachment Lifecycle, Magic Bytes Security & Row-Level Lock Protection

7.1 Attachment Rules & Security

- **Allowed Formats:** `.jpg`, `.jpeg`, `.png`, `.webp`, `.pdf` with size limit ≤ 5 MB per file. - **Magic Bytes Validation** (`validateFileBufferSignature`): Inspects first 12 bytes of uploaded file buffer (e.g. `%PDF-` for PDF, `0xFF 0xD8 0xFF` for JPEG, `0x89 0x50 0x4E 0x47` for PNG, `RIFF...WEBP` for WEBP) to prevent disguised malicious file uploads. - **Active Limit Concurrency Guard:** Post-creation uploads (`POST /api/tickets/:id/attachments`) execute inside a Prisma transaction with row-level locking (`SELECT id FROM "Ticket" WHERE id = ${ticketId} FOR UPDATE`) to enforce max 5 active attachments under concurrent upload requests. - **Soft-Removal with Reason** (`DELETE /api/attachments/:id/soft-remove`): Soft-removes attachment record (`isRemoved: true` , records `removalReason` min 3 chars, and `removedAt` timestamp). - **Stream Blocking:** `GET /api/attachments/:id/download` checks `isRemoved` flag; soft-removed attachments return `403 Forbidden` and block binary file streaming.

7.2 Responsive Visual Screenshots Documentation

All 9 required screenshots were captured using automated Playwright layout tests across Desktop (1280px), Tablet (768px), and Mobile (375px) viewports:

Create Ticket Form Viewport Layouts - Desktop (1280px):

TokTickIT

My Tickets

Create Ticket

Jennifer AndersonEngineeringChange Requester

Create IT Support Ticket

Submit a new IT request for hardware, software, account access, or network support. Fields marked with * are required.

Submitting as: Jennifer Anderson (jennifer.anderson@example.com) — Engineering

Ticket Number (Read-only)

TKT-YYYY-XXXXXX (Auto-generated upon submission)

Ticket Date (Read-only)

2026-09-05 (Today)

Category *

Account and Access

Related System *

Email

Requested Priority *

LOW

MEDIUM

HIGH

URGENT

Summary *

Responsive UI Test (desktop)

28/120

Provide a concise one-line description of your issue (5-120 characters).

Description *

Capturing responsive visual design system inspection screenshot with initial attachment.

88/2000

Describe the problem in detail (10-2000 characters).

Initial Attachments (Optional — Max 5 files)

1 / 5 files

เลือกไฟล์ไม่มีไฟล์ที่เลือก

Allowed formats: .jpg, .jpeg, .png, .webp, .pdf (Max file size: 5 MB per file).

PDFsample-attachment.pdf(0.00 MB)

Remove

Cancel

Submit Ticket

System Health & API Check

Check System Status

Create Ticket Desktop

- Tablet (768px):

Create IT Support Ticket

Submit a new IT request for hardware, software, account access, or network support. Fields marked with * are required.

Submitting as: **Jennifer Anderson** (jennifer.anderson@example.com) — *Engineering*

Ticket Number (Read-only)

TKT-YYYY-XXXXXX (Auto-generated upon su

Ticket Date (Read-only)

2026-09-05 (Today)

Category *

Account and Access

Related System *

Email

Requested Priority *

☐ LOW ☒ MEDIUM ☐ HIGH ☐ URGENT

Summary *

27/120

Responsive UI Test (tablet)

Provide a concise one-line description of your issue (5-120 characters).

Description *

88/2000

Capturing responsive visual design system inspection screenshot with initial attachment.

Describe the problem in detail (10-2000 characters).

Initial Attachments (Optional — Max 5 files)

1 / 5 files

เลือกไฟล์ ไม่มีไฟล์ที่เลือก

Allowed formats: .jpg, .jpeg, .png, .webp, .pdf (Max file size: 5 MB per file).

PDF sample-attachment.pdf (0.00 MB)

Remove

Cancel

Submit Ticket

System Health & API Check

Check System Status

Create Ticket Tablet

- Mobile (375px):

Create IT Support Ticket

Submit a new IT request for hardware, software, account access, or network support. Fields marked with * are required.

Submitting as: **Jennifer Anderson**
(jennifer.anderson@example.com) —
Engineering

Ticket Number (Read-only)

TKT-YYYY-XXXXXX (Auto-generated up

Ticket Date (Read-only)

2026-09-05 (Today)

Category *

Account and Access



Related System *

Email



Requested Priority *

☐ LOW ☒ MEDIUM ☐ HIGH

☐ URGENT

Summary *

27/120

Responsive UI Test (mobile)

Provide a concise one-line description of your issue (5-120 characters).

Description *

88/2000

Capturing responsive visual design system inspection screenshot with initial attachment.

Describe the problem in detail (10-2000 characters).

Initial Attachments (Optional) 1 / 5
— Max 5 files files

เลือกไฟล์ ไม่มีไฟล์ที่เลือก

Allowed formats: .jpg, .jpeg, .png, .webp, .pdf (Max file size: 5 MB per file).

PDF

sample-a...

(0.00
MB)

Remove

Cancel

Submit Ticket

System Health & API Check

Check System Status

My Tickets Viewport Layouts - Desktop (1280px):

TokTickIT

My Tickets

Create Ticket

Jennifer AndersonEngineeringChange Requester

Welcome, Jennifer Anderson👋

Create Ticket

My Tickets — View and track support tickets created by Jennifer Anderson

Search Ticket No or Summary...

All Categories

All Prioritys

All Status

Newest First

Ticket No	Date	Summary	Category	Priority	Status	Files	Action
TKT-2026-000293	Sep 5, 2026	E2E Full Journey Ticket - 1788621886481	Account and Access	HIGH	NEW	1	View
TKT-2026-000292	Sep 5, 2026	Responsive UI Test (mobile)	Account and Access	HIGH	NEW	1	View
TKT-2026-000291	Sep 5, 2026	Responsive UI Test (tablet)	Account and Access	HIGH	NEW	1	View
TKT-2026-000290	Sep 5, 2026	Responsive UI Test (desktop)	Account and Access	HIGH	NEW	1	View
TKT-2026-000289	Sep 5, 2026	E2E Full Journey Ticket - 1788621858470	Account and Access	HIGH	NEW	1	View
TKT-2026-000288	Sep 5, 2026	Responsive UI Test (mobile)	Account and Access	HIGH	NEW	1	View
TKT-2026-000287	Sep 5, 2026	Responsive UI Test (tablet)	Account and Access	HIGH	NEW	1	View
TKT-2026-000286	Sep 5, 2026	Responsive UI Test (desktop)	Account and Access	HIGH	NEW	1	View
TKT-2026-000285	Sep 5, 2026	E2E Full Journey Ticket - 1788621803109	Account and Access	HIGH	NEW	1	View
TKT-2026-000284	Sep 5, 2026	Responsive UI Test (mobile)	Account and Access	HIGH	NEW	1	View

Showing 1 to 10 of 272 tickets

«

1

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27

28

Next

PreviousNext

System Health & API Check

Check System Status

- Tablet (768px):

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requester

Welcome, Jennifer Anderson

My Tickets — View and track support tickets created by Jennifer Anderson

Create Ticket

Search Ticket No or Su

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Newest F

Ticket No	Date	Summary	Category	Priority	Status	Files	Action
TKT-2026-000294	Sep 5, 2026	Responsive UI Test (desktop)	Account and Access	HIGH	NEW	1	View
TKT-2026-000293	Sep 5, 2026	E2E Full Journey Ticket - 1788621886481	Account and Access	HIGH	NEW	1	View
TKT-2026-000292	Sep 5, 2026	Responsive UI Test (mobile)	Account and Access	HIGH	NEW	1	View
TKT-2026-000291	Sep 5, 2026	Responsive UI Test (tablet)	Account and Access	HIGH	NEW	1	View
TKT-2026-000290	Sep 5, 2026	Responsive UI Test (desktop)	Account and Access	HIGH	NEW	1	View
TKT-2026-000289	Sep 5, 2026	E2E Full Journey Ticket - 1788621858470	Account and Access	HIGH	NEW	1	View
TKT-2026-000288	Sep 5, 2026	Responsive UI Test (mobile)	Account and Access	HIGH	NEW	1	View
TKT-2026-000287	Sep 5, 2026	Responsive UI Test (tablet)	Account and Access	HIGH	NEW	1	View
TKT-2026-000286	Sep 5, 2026	Responsive UI Test (desktop)	Account and Access	HIGH	NEW	1	View
TKT-2026-000285	Sep 5, 2026	E2E Full Journey Ticket - 1788621803109	Account and Access	HIGH	NEW	1	View

Showing 1 to 10 of 273 tickets

< Previous

12345678910111213141516171819202122232425262728

Next >

System Health & API Check

Check System Status

My Tickets Tablet

- Mobile (375px):

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requester

Welcome, Jennifer Anderson

My Tickets — View and track support tickets created by Jennifer Anderson

Create Ticket

Search Ticket No or Summary...

All CategorysAll PrioritysAll StatussNewest First

TKT-2026-000295

NEW

Responsive UI Test (tablet)

Account and AccessHIGH1 files

Sep 5, 2026View Details

TKT-2026-000294

NEW

Responsive UI Test (desktop)

Account and AccessHIGH1 files

Sep 5, 2026View Details

TKT-2026-000293

NEW

E2E Full Journey Ticket - 1788621886481

Account and AccessHIGH1 files

Sep 5, 2026View Details

TKT-2026-000292

NEW

Responsive UI Test (mobile)

Account and AccessHIGH1 files

Sep 5, 2026View Details

TKT-2026-000291

NEW

Responsive UI Test (tablet)

Account and AccessHIGH1 files

Sep 5, 2026View Details

TKT-2026-000290

NEW

Responsive UI Test (desktop)

Account and AccessHIGH1 files

Sep 5, 2026View Details

TKT-2026-000289

NEW

E2E Full Journey Ticket - 1788621858470

Account and AccessHIGH1 files

Sep 5, 2026View Details

TKT-2026-000288

NEW

TKT-2026-000286

NEW

Responsive UI Test (mobile)

Account and AccessHIGH1 files

Sep 5, 2026View Details

TKT-2026-000287

NEW

Responsive UI Test (tablet)

Account and AccessHIGH1 files

Sep 5, 2026View Details

TKT-2026-000286

NEW

Responsive UI Test (desktop)

Account and AccessHIGH1 files

Sep 5, 2026View Details

Showing 1 to 10 of 274 tickets

<Previous

12345678910111213141516171819202122232425262728

Next>

System Health & API Check

Check System Status

Ticket Detail Viewport Layouts - Desktop (1280px):

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Engineering

Change Requester

My Tickets / Ticket Details (TKT-2026-000294)

Ticket Date: Sep 5, 2026, 10:25 PM

Priority: HIGHStatus: NEW

TKT-2026-000294

Development Requester

Jennifer Anderson

jennifer.anderson@example.com

Category

Account and Access

Related System

Email

Ticket Summary

Responsive UI Test (desktop)

Description

Capturing responsive visual design system inspection screenshot with initial attachment.

File Attachments

Active Attachments: 1 / 5

Upload New Attachment

เลือกไฟล์

ไม่มีไฟล์ที่เลือก

Upload Attachment

Allowed formats: .jpg, .jpeg, .png, .webp, .pdf (Max file size: 5 MB)

Active Files (1)

File Name	Size	Uploaded Date	Actions
sample-attachment.pdf application/pdf	37 Bytes	Sep 5, 2026, 10:25 PM	Download Soft Remove

Back to My Tickets List

System Health & API Check

Check System Status

Ticket Detail Desktop

- Tablet (768px):

My Tickets / Ticket Details (TKT-2026-000295)

Ticket Date: Sep 5, 2026, 10:25 PM

Priority: HIGH

Status: NEW

TKT-2026-000295

Development Requester

Jennifer Anderson

jennifer.anderson@example.com

Category

Account and Access

Related System

Email

Ticket Summary

Responsive UI Test (tablet)

Description

Capturing responsive visual design system inspection screenshot with initial attachment.

File Attachments

Active Attachments: 1 / 5

Upload New Attachment

เลือกไฟล์

ไม่มีไฟล์ที่เลือก

Upload Attachment

Allowed formats: .jpg, .jpeg, .png, .webp, .pdf (Max file size: 5 MB)

Active Files (1)

File Name	Size	Uploaded Date	Actions
sample-attachment.pdf application/pdf	37 Bytes	Sep 5, 2026, 10:25 PM	Download Soft Remove

[« Back to My Tickets List](#)

System Health & API Check

Check System Status

Ticket Detail Tablet

- Mobile (375px):

 My Tickets

/ Ticket Details (TKT-2026-000296)

Ticket Date: Sep 5, 2026, 10:25 PM

TKT-2026-000296

Priority: HIGH

Status: NEW

Development Requester

Jennifer Anderson

jennifer.anderson@example.com

Category

Account and Access

Related System

Email

Ticket Summary

Responsive UI Test (mobile)

Description

Capturing responsive visual design system inspection screenshot with initial attachment.



File Attachments

Active Attachments: 1 / 5

Upload New Attachment

เลือกไฟล์

ไม่มีไฟล์ที่เลือก

Allowed formats: .jpg, .jpeg, .png, .webp, .pdf (Max file size: 5 MB)

Upload Attachment

Active Files (1)

File Name	Size	Uploaded Date	
sample-attachment.pdf application/pdf	37 Bytes	Sep 5, 2026, 10:25 PM	Dov

« Back to My Tickets List

System Health & API Check

Check System Status

Answer Part 8: Complete Automated Test Execution & Traceability Matrix

8.1 Acceptance-Criterion Traceability Matrix

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Acceptance Criterion	Description	Validating Test Files	Result
AC-01	Create Ticket with Ticket Number & Ticket Date	create-ticket.api.test.ts , CreateTicket.test.tsx , requester-ticket-flow.spec.ts	Pass
AC-02	Requester Selector required when unselected	RequesterSelect.test.tsx , requester-ticket-flow.spec.ts	Pass
AC-03	My Tickets owned tickets filtering	my-tickets.api.test.ts , MyTickets.test.tsx , requester-ticket-flow.spec.ts	Pass
AC-04	Cross-Requester Ticket Access Blocked (403 Forbidden)	ticket-detail.api.test.ts , attachments.api.test.ts , requester-ticket-flow.spec.ts	Pass
AC-05	Soft removal with reason & blocked download	attachments.api.test.ts , AttachmentSection.test.tsx , requester-ticket-flow.spec.ts	Pass
AC-06	Max 5 active attachments limit	create-ticket.api.test.ts , attachments.api.test.ts	Pass
AC-07	File size & type restriction validation	create-ticket.api.test.ts , CreateTicket.test.tsx , AttachmentSection.test.tsx	Pass
AC-08	Search term in My Tickets	my-tickets.api.test.ts , MyTickets.test.tsx , requester-ticket-flow.spec.ts	Pass
AC-09	Ticket Date displayed on screens	create-ticket.api.test.ts , CreateTicket.test.tsx , TicketDetail.test.tsx	Pass
AC-10	Form values preserved on submission failure	CreateTicket.test.tsx	Pass

8.2 Summary of Test Execution

- **Server Unit & Integration Tests:** 54 / 54 passed (`npm run test --prefix server`) - **Client Unit & Component Tests:** 41 / 41 passed (`npm run test --prefix client`) - **Playwright E2E Tests:** 4 / 4 passed (`npx playwright test`) - **Total Test Suites:** 100% Green across all 6 testing levels.

Answer Part 9: Peer Review Resolution Log & AI Pair Programming Reflection

9.1 Peer Review Resolution Log

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PR #	Branch	Description	Reviewer (<code>Imaybelgrace</code>) Comment & Resolution	Status
#23	<code>feature/5-spec-and-tests</code>	Sprint Specifications & Test Plan	Added Appendix B Section 7 Known Limitations and Status Code 403 Forbidden consistency.	Merged
#25	<code>feature/6-db-schema-seed</code>	Database Schema & Seed Data	Implemented Prisma upsert queries for idempotent seed execution.	Merged
#27	<code>feature/7-requester-context</code>	Requester Context Selector	Added persistent localStorage context and header synchronization.	Merged
#29	<code>feature/8-create-ticket-api</code>	Create Ticket API Endpoint	Implemented atomic sequence generator for <code>TKT-YYYY-XXXXXX</code> format.	Merged
#30	<code>feature/9-create-ticket-ui</code>	Create Ticket UI Form	Designed Zen Green form components with asterisks and field validation text.	Merged
#31	<code>feature/10-my-tickets-api</code>	My Tickets API Endpoint	Added multi-select filters, search query, sorting, and pagination.	Merged
#32	<code>feature/11-my-tickets-ui</code>	My Tickets UI View	Implemented debounced search, responsive layout, and mobile cards view.	Merged
#33	<code>feature/12-ticket-detail</code>	Ticket Detail & Ownership Guard	Enforced <code>requesterId</code> query/header sync and ownership check (<code>403 Forbidden</code>).	Merged
#34	<code>feature/13-attachment-lifecycle</code>	Attachment Stream & Removal	Added magic bytes buffer check, row-level locking transaction, and soft removal.	Merged
#35	<code>feature/14-qa-release</code>	QA, Initial Attachments &	Added initial attachments on POST <code>/api/tickets</code> with transaction rollback compensation, expanded Playwright	Open

PR #	Branch	Description	Reviewer (Imaybelgrace)	Comment & Resolution	Status
		Final Release		E2E test, regenerated 9 responsive screenshots, and compiled final deliverable report.	

9.2 AI Pair Programming Reflection

Working with the AI pair programmer under Spec DD method: 1. **Spec DD Precision:** Writing clear specification markdown files prior to coding ensured zero architectural misalignment on data isolation boundaries, attachment soft-removal rules, and magic bytes validation. 2. **Transaction & Compensation Design:** The AI assisted in constructing robust compensation logic for initial attachment file uploads, ensuring zero disk leak when ticket creation transactions fail. 3. **Comprehensive Test Coverage:** The combination of unit, integration, component, and Playwright E2E tests verified that all 10 Acceptance Criteria were 100% covered and resilient against regression.